

4(a)	$E = \frac{1}{2}mv^2$	C1
	$p = mv$	C1
	$m = 0.37^2 / (2 \times 0.30) \text{ or } 0.37 / 1.6 \text{ or } (0.30 \times 2) / 1.6^2$ $= 0.23 \text{ kg}$	A1
4(b)	$0.37 - 0.65 = -0.13 - p$ $p = 0.15 \text{ kg m s}^{-1}$	A1
4(c)	$7.7 = (0.13 + 0.37) / (\Delta)t$ or $7.7 = (0.65 - 0.15) / (\Delta)t$	C1
	time = 0.065 s	A1